# Pratik Pakhale

Computer Science and Engineering
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## Education

• **Indian Institute of Information Technology, Dharwad** *2021-2025*
*Bachelor of Technology in Computer Science and Engineering*

## Experience

• **Raccoon AI** *Dec '24 - Present*
*Founding Engineer (prev. Intern)*
– Shipped **Elemetry, a cache engine** that stores browser actions in an in-memory vector graph database, leveraging browsing history to accelerate repetitive tasks.
– Designed dual-model (ephemeral events vs. persisted blocks) **real-time streaming pipeline** for frontend rendering of realtime agentic work.
– Built a text-to-video motion-graphics system with a **React rendering engine**, director's-board approval flow, parallelized frame-by-frame export on serverless GPUs, and a full audio toolkit.
– Also shipped canvas editor, raccoon-inspector, worked heavily on chat app's state management and performance.

• **Google Summer of Code | Pecan Project** *Summer 2026 & 2024*
*Mentor & Contributor*
– **Mentor (2026):** Selected as a **GSoC mentor** for *LLM-Assisted Extraction of Agronomic and Ecological Experiments into Structured Data*, guiding a contributor through project scoping, technical design, and delivery.
– **Contributor (2024):** Integrated a **JSON schema parser** into the FieldActivity R Shiny application, enabling validation and storage of management-event data.

• **Affect Aware System and QGIS LMS Platform | IIT Bombay** *Summer 2023*
*Research Intern and Developer* IIT Bombay
– Developed an **advanced learning system** utilizing video bots for enhanced student engagement.
– Implemented an adaptive feedback platform using React & ExpressJS, a Flask server running the deep learning model, and improved learning outcomes.
– Built a software that can be executed locally using **Docker** or through an **Electron installation**.
– Working on GeoGuru, a Learning Management System that incorporates a geographical information system. Working as the **lead backend developer** to build scalable APIs.

## Personal Projects

– **Rewind** *2022*
*A social media platform powered with generative AI technologies solely built for music lovers.*
∗ Primarily developed an AI-driven social media platform catered to music enthusiasts using TypeScript Node, React, and Python.
∗ Spearheaded the design and implementation of a **voice bot for enhanced website interaction**, including post-creation and advanced search functionalities.

– **Event Management** *2021*
*The complete event management website for club events from registering to ticketing.*
∗ Solely created a one-stop online platform for managing club events, including registration and ticketing, using MERN stack, TailwindCSS, and Netlify Lambda.
∗ Successfully maintained errorless operation handling over 100k API calls.

## Technical Skills and Interests

**Languages**: TypeScript, Python, Go, Java, C, C++, R, HTML, CSS
**Tools**: Docker, GitHub Actions, AWS and CI workflows
**Frameworks**: Nextjs, FastAPI
**Databases**: Vector, Graph, SQL, and No-SQL

## Extras

– **Lead:** Multiple Institute Websites (Full Next.js Turborepo) - iiitdwd.ac.in
– **President:** Velocity, Web and App Development Club
– **Winner and Student Leader:** Postman Inc. Global Hackathon
– **Campus Rep:** Buildspace, Nights & Weekends